

0	Create a Satellite Digital Twin (SDT)
REQ-CORE-1.0	
1.1	The SDT shall be able to process inputs with expected outputs
1.1.1	The SDT shall use tools for implementation
1.1.1.1	The SDT shall be coded within Python
1.1.1.2	The SDT shall be created in GitLab
1.1.1.3	The SDT shall be implemented by Cerux for C3
1.1.1.4	The SDT shall be contained in Docker
1.1.1.4.1	The SDT shall use Kubernetes to run Docker
1.1.1.5	Data shall be developed within amazon workspace
1.1.1.6	The SDT shall contain a visualization board
1.1.1.7	The SDT shall simulate orbital mechanics
1.1.1.8	All data shall be stored within a databank
1.1.2	The SDT shall provide command and control data
1.1.2.1	C2 data shall provide movement commands
1.1.2.1.1	C2 data shall provide rotational movement
1.1.2.1.2	C2 data shall reposition the satellite
1.1.2.1.3	C2 data shall provide momentum instruction
1.1.3	The SDT shall simulate environmental impacts to the satellite
1.1.3.1	The SDT shall simulate weather conditions
1.1.3.2	The SDT shall simulate unknown objects
1.1.3.2.1	Simulation data shall consider space debris
1.1.3.2.2	Simulation data shall consider unidentified technologies
1.1.4	All data shall be in metric units
REQ-INT-1.0	
1.1	The SDT shall integrate with Auria applications
1.1.1	The SDT receives master satellite scheduling
1.1.1.1	Satellite shall emit a timestamp
1.1.1.2	Satellite shall emit event_ID
1.1.2	The SDT shall process telemetry data to determine system health
1.1.2.1	The telemetry data shall contain orbital data
1.1.2.2	The telemetry data shall contain orientation data
1.1.2.3	The telemetry data shall contain battery levels
1.1.2.4	The telemetry data shall contain battery health
1.1.2.5	The telemetry data shall contain temperature data
1.1.2.5.1	The telemetry data shall contain temperature data for bus
1.1.2.5.2	The telemetry data shall contain temperature data for payload
1.1.2.5.3	The telemetry data shall contain temperature data for components
1.1.3	The SDT shall contain different options for data compatibility
1.1.3.1	Data shall be consumed to TeRMS
1.1.3.2	Data shall be consumed by MFPA
1.1.3.2.1	MFPA shall deconflict mission planning
REQ-APP-1.0	
1.1	The SDT shall be applicable for different use cases

1.1.1	The SDT shall simulate mission readiness of combat systems
1.1.1.1	The SDT shall implement cybersecurity testing
1.1.1.2	The SDT shall simulate cybersecurity vulnerability
1.1.1.3	The SDT shall highlight future technology development
1.1.2	The SDT shall monitor environmental effects on different systems
1.1.2.1	Sensors shall detect solar flares
1.1.2.2	Sensors shall detect radiation
1.1.2.3	Sensors shall detect atmospheric disturbances
1.1.2.4	Sensors shall detect geomagnetic storms
1.1.2.5	Signal and network resilience for environmental conditions
1.1.3	The SDT shall provide operator training for satellite use
1.1.3.1	Simulated proximity events for operator mission rehearsal
1.1.3.2	Simulated multidomain training for operator rehearsal
1.1.3.3	Simulated C2 training for operator rehearsal
1.1.3.3.1	Simulated communication training for operator rehearsal
1.1.3.3.2	Simulated network training for operator rehearsal
1.1.3.3.3	Shall be applicable to Forgepoint operator training
1.1.4	The SDT shall provide C3 system development
1.1.4.1	The SDT shall contain visual system health indicators
1.1.4.1.1	Green visual indicators shall correspond to healthy systems
1.1.4.1.2	Yellow visual indicators shall correspond to at risk systems
1.1.4.1.3	Orange visual indicators shall correspond to critical systems
1.1.4.1.4	Red visual indicators shall correspond to failed systems